



Part I

Lab Assignment-2 (Introduction to Programming Structure)

Department: CSE
(22CPP209)

Subject: OOAD Lab

Objective: *Understand the structure of a C++ program header files, the compilation and linking process, stringstream data type, function overloading and functions templates.*

Question 1: Create a header file named **arithmetic.h** that contains declarations for arithmetic operations such as- addition, subtraction, multiplication etc. Provide the definitions of the function declarations of header in a source file **arithmetic.cpp**. Finally, write a program that includes the **arithmetic.h** header file and uses the arithmetic functions. Understand and implement header guards in **arithmetic.h** to prevent multiple inclusions of the same header file.

Question 2: Using algebra.cpp, and main.cpp, show—through a single experiment—how static and shared linking differ. First, compile the two implementation files into object code, archive them into a static library named **libmath.a**, and link that library to build an executable called **demo_static**. Then repeat the process with position-independent code to create a shared object **libmath.so**, link it to form **demo_shared**, and ensure the shared library is located automatically at run time. Finally, run exactly one command on each executable that confirms its linkage type, record the resulting output, and observe when you would favour a static build over a dynamic one (and vice-versa) with respect to size, distribution, and update flexibility.